Problem Set 3

Applied Stats/Quant Methods 1

Due: November 12, 2026

Instructions

- Please show your work! You may lose points by simply writing in the answer. If the problem requires you to execute commands in `R`, please include the code you used to get your answers. Please also include the `.R` file that contains your code. If you are not sure if work needs to be shown for a particular problem, please ask.
- Your homework should be submitted electronically on GitHub.
- This problem set is due before 23:59 on Thursday November 12, 2026. No late assignments will be accepted.

In this problem set, you will run several regressions and create an added variable plot in `R` using the `incumbents_subset.csv` dataset. Include all of your code.

Question 1

We are interested in knowing how the difference in campaign spending between incumbent and challenger affects the incumbent's vote share.

1. Run a regression where the outcome variable is `voteshare` and the explanatory variable is `difflog`.

```r
model_voteshare <- lm(voteshare ~ difflog, data = inc.sub)
model_voteshare

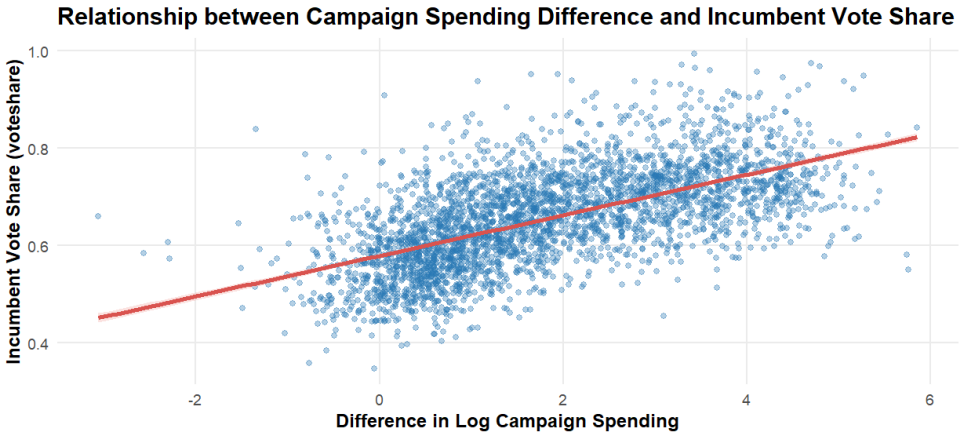
# Display regression results
summary(model_voteshare)
```

Interpretation: The intercept is 0.579031 (predicted incumbent vote share when spending difference is zero is approx. 57.90%). The coefficient for difflog is 0.041666, indicating that a one-unit increase in the spending difference is associated with an increase of approximately 4.17 percentage points in the incumbent vote share. The

p-value ($< 2.2e-16$) indicates statistical significance, and $R^2 \approx 0.3673$. My model explains 36% of the variation in vote share (voteshare) across the observations.

2. Make a scatterplot of the two variables and add the regression line.

```
ggplot(inc.sub, aes(x = difflog, y = voteshare)) +
  geom_point(color = "#2878B5", alpha = 0.35, size = 1.5) +
  geom_smooth(method = "lm", se = TRUE,
              color = "#D9534F", fill = "#F4B6B2",
              linewidth = 1.2) +
  labs(
    title = "Relationship between Campaign Spending Difference and
   Incumbent Vote Share",
    x = "Difference in Log Campaign Spending",
    y = "Incumbent Vote Share (voteshare)"
  ) +
  theme_minimal(base_size = 12) +
  theme(
    plot.title = element_text(face = "bold", size = 15),
    plot.subtitle = element_text(size = 10, color = "grey40"),
    axis.title = element_text(face = "bold"),
    panel.grid.minor = element_blank(),
    plot.caption = element_text(color = "grey50"),
```



Interpretation: There is a positive association in this linear relationship between the two variables. As the campaign spending increases so does incumbent vote share.

3. Save the residuals of the model in a separate object.

```
residual_model_voteshare <- residuals(model_voteshare)
head(residual_model_voteshare)
```

4. Write the prediction equation.

   Prediction Equation:

$$\widehat{\text{voteshare}} = 0.579031 + 0.041666 \times \text{difflog}$$

```
# Predicted incumbent voter-share = 0.579031 + 0.041666 (difflog)
coef(model_voteshare)
```

   Interpretation: When the difference in log campaign spending is zero (difflog = 0), the model predicts an incumbent vote share of 0.579, or about 57.9%. For every one-unit increase in the difference in log campaign spending (difflog), the incumbent's predicted vote share increases by 0.041666 (or approximately 4.17 percentage points).

Question 2

We are interested in knowing how the difference between incumbent and challenger's spending and the vote share of the presidential candidate of the incumbent's party are related.

1. Run a regression where the outcome variable is `presvote` and the explanatory variable is `difflog`.

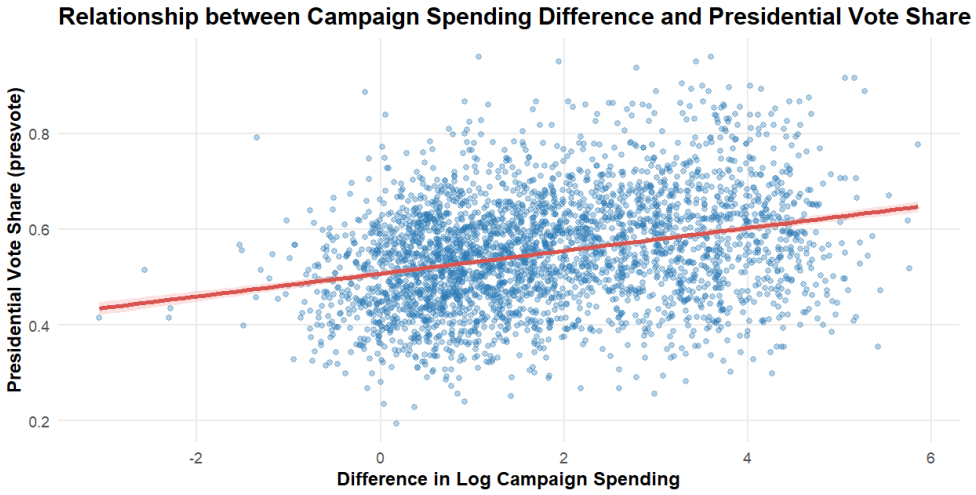
```r
model_presvote <- lm(presvote ~ difflog, data = inc.sub)
model_presvote

# Display regression results
summary(model_presvote)
```

Interpretation: The intercept is 0.507583 (predicted incumbent vote share when spending difference is zero is approx. 50.76%). The coefficient for difflog is 0.023837, indicating that a one-unit increase in the spending difference is associated with an increase of approximately 2.38 percentage points in the incumbent vote share. The p-value ($< 2e-16$) indicates statistical significance, and $R^2 \approx 0.0880$. My model explains 8.8% of the variation in vote share (presvote) across the observations.

2. Make a scatterplot of the two variables and add the regression line.

```
ggplot(inc.sub, aes(x = difflog, y = presvote)) +
  geom_point(color = "#2878B5", alpha = 0.35, size = 1.5) +
  geom_smooth(method = "lm", se = TRUE,
              color = "#D9534F", fill = "#F4B6B2",
              linewidth = 1.2) +
  labs(
    title = "Relationship between Campaign Spending Difference and
    Presidential Vote Share",
    x = "Difference in Log Campaign Spending",
    y = "Presidential Vote Share (presvote)"
  ) +
  theme_minimal(base_size = 12) +
  theme(
    plot.title = element_text(face = "bold", size = 15),
    plot.subtitle = element_text(size = 10, color = "grey40"),
    axis.title = element_text(face = "bold"),
    panel.grid.minor = element_blank(),
    plot.caption = element_text(color = "grey50"),
```



Interpretation: There is a positive association in this linear relationship between the two variables. As the campaign spending increases so does the presidential vote share.

3. Save the residuals of the model in a separate object.

```
residual_model_presvote <- residuals(model_presvote)
```

4. Write the prediction equation.

Prediction Equation:

$$\widehat{\text{presvote}} = 0.50758 + 0.02384 \times \text{difflog}$$

```
# Presidential candidate of the incumbent's party = 0.50758333 +
    0.02383723 (difflog)
coef(model_presvote)
```

Interpretation: When the difference in log campaign spending is zero (difflog = 0), the model predicts a presidential vote share of 0.508, or about 50.76%. For every one-unit increase in the difference in log campaign spending (difflog), the incumbent's predicted presidential vote share increases by 0.0238 (or approximately 2.38 percentage points).

Question 3

We are interested in knowing how the vote share of the presidential candidate of the incumbent's party is associated with the incumbent's electoral success.

1. Run a regression where the outcome variable is `voteshare` and the explanatory variable is `presvote`.

```r
model_voterpres <- lm(voteshare ~ presvote, data = inc.sub)
model_voterpres

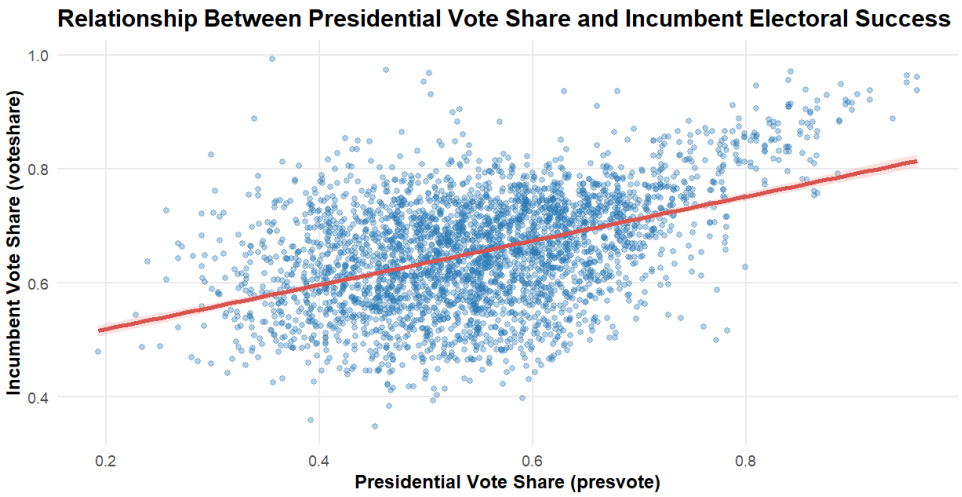
# Display regression results
summary(model_voterpres)
```

Interpretation: The intercept is 0.441330 (predicted incumbent vote share when presidential vote share is zero is approx. 44.13%). The coefficient for presvote is 0.388018, indicating that a one-unit increase in presidential vote share is associated with an increase of approximately 38.80 percentage points in the incumbent vote share. The p-value ($< 2e-16$) indicates statistical significance, and $R^2 \approx 0.2058$. My model explains 20.58% of the variation in vote share (voteshare) across the observations.

2. Make a scatterplot of the two variables and add the regression line.

```r
ggplot(inc.sub, aes(x = presvote, y = voteshare)) +
  geom_point(color = "#2878B5", alpha = 0.35, size = 1.5) +
  geom_smooth(method = "lm", se = TRUE,
              color = "#D9534F", fill = "#F4B6B2",
              linewidth = 1.2) +
  labs(
    title = "Relationship Between Presidential Vote Share and Incumbent
    Electoral Success",
    x = "Presidential Vote Share (presvote)",
    y = "Incumbent Vote Share (voteshare)"
  ) +
  theme_minimal(base_size = 12) +
  theme(
    plot.title = element_text(face = "bold", size = 15),
    plot.subtitle = element_text(size = 10, color = "grey40"),
    axis.title = element_text(face = "bold"),
    panel.grid.minor = element_blank(),
    plot.caption = element_text(color = "grey50"),
    plot.margin = margin(15, 15, 15, 15)
  )
```

Interpretation: There is a positive association in this linear relationship between the two variables. As the presidential vote share increases so does the incumbent vote share.



3. Write the prediction equation.

   Prediction Equation:

$$\widehat{\text{voteshare}} = 0.441330 + 0.388018 \times \text{presvote}$$

```r
# Presidential vote share and incumbent electoral success = 0.4413299 +
    0.3880184(presvote)
coef(model_voterpres)
```

   Interpretation: When the presidential vote share is zero (presvote = 0), the model predicts an incumbent vote share of 0.441, or about 44.13%. For every one-unit increase in the presidential vote share (presvote), the incumbent's predicted vote share increases by 0.3880 (or approximately 38.80 percentage points).

Question 4

The residuals from part (a) tell us how much of the variation in `voteshare` is *not* explained by the difference in spending between incumbent and challenger. The residuals in part (b) tell us how much of the variation in `presvote` is *not* explained by the difference in spending between incumbent and challenger in the district.

1. Run a regression where the outcome variable is the residuals from Question 1 and the explanatory variable is the residuals from Question 2.

```r
model_residuals <- lm(residual_model_voteshare ~ residual_model_presvote)

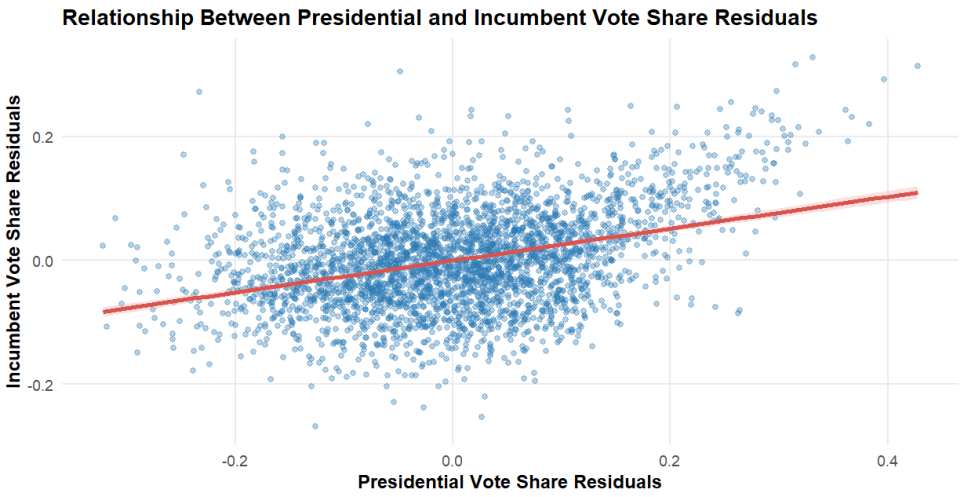
# Display regression results
summary(model_residuals)
```

Interpretation: The intercept is $-5.934\text{e}-18$ (predicted residual incumbent vote share when residual presidential vote share is zero is approx. 0.00%). The coefficient for residual_model_presvote is 0.2569, indicating that a one-unit increase in the residual presidential vote share is associated with an increase of approximately 25.69 percentage points in the residual incumbent vote share. The p-value ($< 2\text{e}{-16}$) indicates statistical significance, and $R^2 \approx 0.1300$. My model explains 13.00% of the variation in residual vote share across the observations.

2. Make a scatterplot of the two residuals and add the regression line.

```r
ggplot(inc.sub, aes(x = residual_model_presvote,
                    y = residual_model_voteshare)) +
  geom_point(color = "#2878B5", alpha = 0.35, size = 1.5) +
  geom_smooth(method = "lm", se = TRUE,
              color = "#D9534F", fill = "#F4B6B2",
              linewidth = 1.2) +
  labs(
    title = "Relationship Between Presidential and Incumbent Vote Share Residuals",
    x = "Presidential Vote Share Residuals",
    y = "Incumbent Vote Share Residuals",
  ) +
  theme_minimal(base_size = 12) +
  theme(
    plot.title = element_text(face = "bold", size = 14),
    plot.subtitle = element_text(size = 10, color = "grey40"),
    axis.title = element_text(face = "bold"),
    panel.grid.minor = element_blank(),
    plot.caption = element_text(color = "grey50"),
    plot.margin = margin(15, 15, 15, 15)
  )
```

Interpretation: There is a positive association in this linear relationship between the two variables. As the presidential vote share increases so does the incumbent vote share.



3. Write the prediction equation.

   Prediction Equation:

$$\widehat{\text{res_voteshare}} = -5.934\text{e}-18 + 0.256877 \times \text{res_presvote}$$

```
# Residual model voteshare = -5.934078e-18 + 0.2568770(residual_model_
    presvote)
coef(model_residuals)
```

   Interpretation: When the residual presidential vote share is zero (residual_model_presvote = 0), the model predicts a residual incumbent vote share of $-5.934\text{e}-18$, or about 0.00%. For every one-unit increase in the residual presidential vote share (residual_model_presvote), the incumbent's predicted residual vote share increases by 0.2569 (or approximately 25.69 percentage points).

Question 5

What if the incumbent's vote share is affected by both the president's popularity and the difference in spending between incumbent and challenger?

1. Run a regression where the outcome variable is the incumbent's `voteshare` and the explanatory variables are `difflog` and `presvote`.

```r
model_multiple <- lm(voteshare ~ difflog + presvote, data = inc.sub)

# Display regression results
summary(model_multiple)
```

   Interpretation: The intercept is 0.4486 (predicted incumbent vote share when both the spending difference and presidential vote share are zero is approx. 44.86%). The p-values ($< 2e-16$) indicate statistical significance, and $R^2 \approx 0.4496$. The model explains approximately 44.96% of the variation in incumbent vote share across the observations.

2. Write the prediction equation.

   Prediction Equation:

$$\widehat{\text{voteshare}} = 0.4486442 + 0.0355431 \times \text{difflog} + 0.2568770 \times \text{presvote}$$

   Interpretation: The coefficient for difflog is 0.0355, indicating that a one-unit increase in the spending difference is associated with an increase of approximately 3.55 percentage points in the incumbent vote share, holding presidential vote share constant. The coefficient for presvote is 0.2569, indicating that a one-unit increase in presidential vote share is associated with an increase of approximately 25.69 percentage points in the incumbent vote share, holding spending difference constant.

3. What is it in this output that is identical to the output in Question 4? Why do you think this is the case?

   The coefficient for `presvote` (0.2569) in Question 5 is identical to the coefficient of the residual regression in Question 4. This is a direct consequence of the **Frisch–Waugh–Lovell theorem** and the concept of statistical control. By removing the effect of difflog (the difference in spending between incumbent and challenger) from both incumbent vote share and presidential vote share, we effectively control for the influence of spending. The R-squared value and the coefficient for difflog are different.